W04–W05 — Combined Research Progress Report

Project: Advanced Chooser Option Pricing with Real-World Data & Machine Learning
Underlying: JPM | **Historical valuation date:** December 31, 2024
Research update: November 4, 2026
Scope: Heston Monte Carlo, exploratory market frictions, and ML volatility forecasting

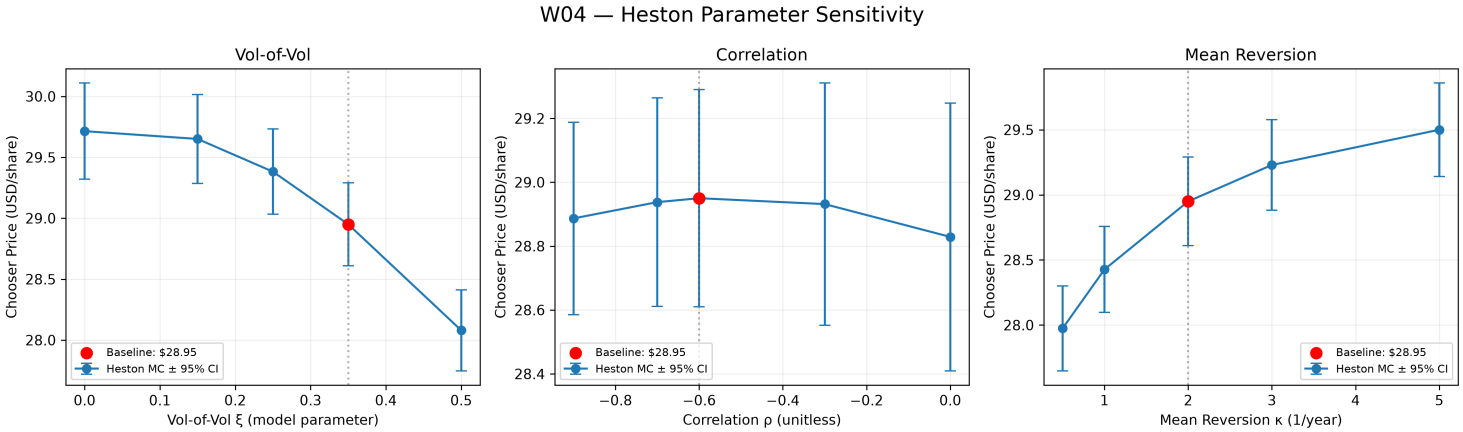
1. Executive Summary

- **W04:** Implemented a Heston stochastic-volatility Monte Carlo benchmark for a Simple Chooser Option, with numerical validation and illustrative transaction-cost scenarios.
- **W05:** Trained a Random Forest to forecast future 20-trading-day realized volatility and compared it with a historical-volatility baseline.
- **Pending:** Market calibration, dynamic hedging costs, and full direct-pricing ML training; these are not claimed as completed.

2. W04 — Heston Monte Carlo

Method: Correlated stock/variance shocks; full-truncation Euler variance scheme; discounted European call + **adjusted put leg** (expiring at the choice date). Heston parameters are **illustrative, not market-calibrated**: mean reversion $\kappa = 2$, vol-of-vol $\xi = 0.35$, correlation $\rho = -0.60$, with $v_0 = \theta = \sigma_{20d}^2$.

Numerical result	Value
W03 dividend-adjusted BSM Chooser reference	**$29.8290/share**
Heston Monte Carlo fair value (100,000 paths)	**$28.9518/share**
Monte Carlo standard error	**$0.0770/share**
95% Monte Carlo confidence interval	**[$28.8010, $29.1027]/share**
Constant-variance (GBM) consistency check	**Passed**; 0.55 standard errors from BSM
Path-count experiment	SE decreased from **0.3488** (5K) to **0.0770** (100K)
Time-step experiment	Estimates had overlapping confidence intervals; further bias testing needed



Heston Parameter Sensitivity

Chart interpretation: Within the tested *illustrative* parameter ranges, higher vol-of-vol lowered the estimated Chooser value and faster mean reversion raised it. The estimated effect of changing the price–variance correlation was relatively modest compared with Monte Carlo uncertainty. These patterns are scenario-specific, not general monotonicity claims.

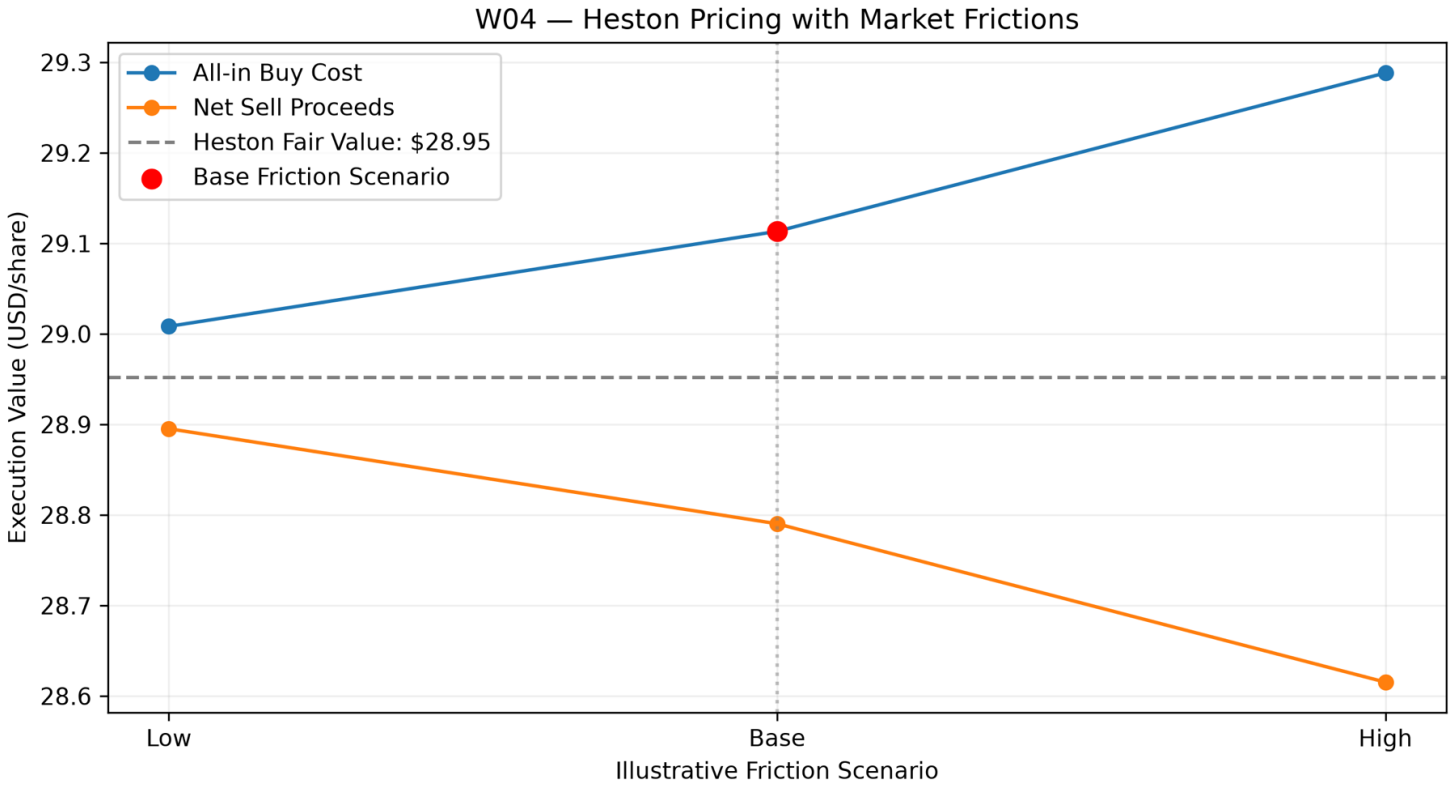
Numerical limitation: The confidence interval measures Monte Carlo sampling error, not discretization bias or Heston parameter uncertainty. The BSM line is a *comparison benchmark*, not the Heston convergence target.

3. Market Frictions — Exploratory Execution Costs

We added a **transaction-level execution-cost layer** to the frictionless Heston fair value, using assumed full bid–ask spreads, additional adverse slippage, and a $0.65 fee per hypothetical 100-share contract.

```
Cost per share = half-spread + slippage + fee/100
All-in buy cost = Heston fair value + cost per share
Net sell proceeds = Heston fair value − cost per share
```

Scenario	Full spread (/share)	$Slippage(/share)$	Buy ($/share)	$Sell(/share)$	Round trip ($/share)
Low	0.10	0.00	29.0083	28.8953	0.1130
Base	**0.25**	**0.03**	**29.1133**	**28.7903**	**0.3230**
High	0.50	0.08	29.2883	28.6153	0.6730



Illustrative Market Frictions

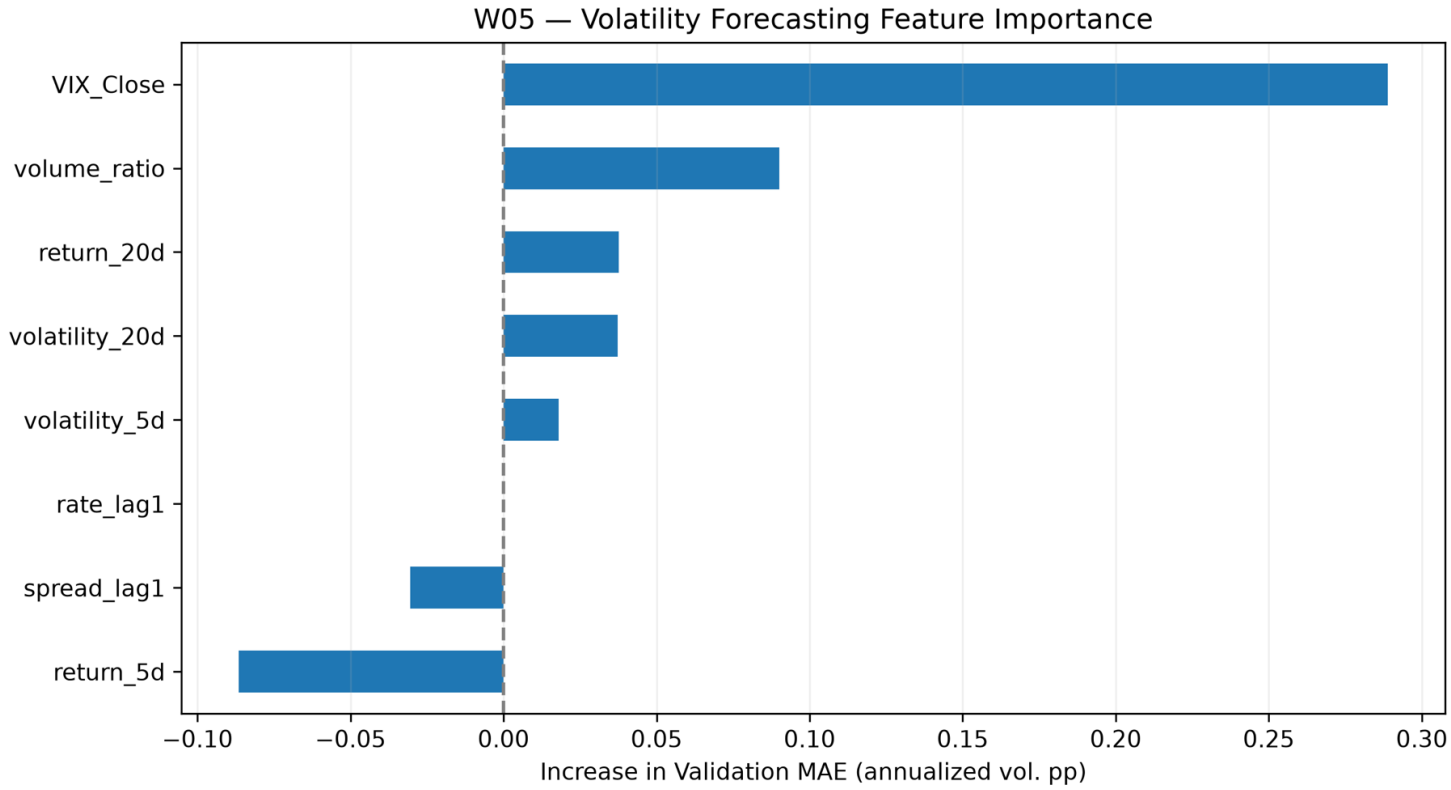
Chart interpretation: Higher assumed frictions widen the gap between all-in buying cost and net selling proceeds; **the frictionless theoretical value remains $28.9518/share**. These inputs are hypothetical, **not observed Chooser quotes**. Dynamic delta-hedging costs are not modeled.

4. W05 — Future Volatility Forecasting

Target: JPM future 20-trading-day realized volatility, annualized.
Inputs: Eight historical market features, including JPM returns, rolling volatility, volume, VIX, and lagged Treasury indicators.
Models: Historical 20-day volatility baseline vs. Random Forest.
Chronological split: Train 2018–2022 (**1,218**), validation 2023 (**230**), test 2024 (**232**). Target windows crossing period boundaries were excluded.

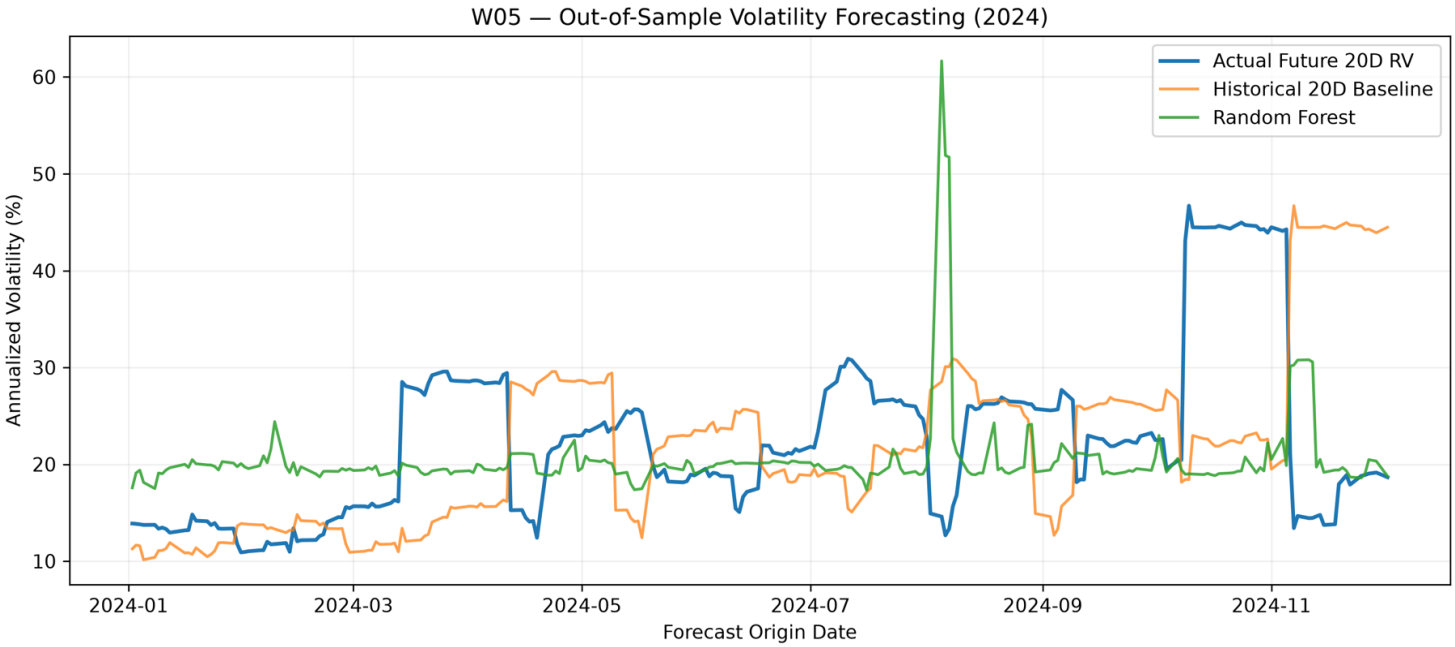
Period	Model	MAE (vol. pp) ↓	RMSE (vol. pp) ↓
Validation	Historical 20D	**6.081**	8.455
Validation	Random Forest	6.201	**8.143**
Test	Historical 20D	9.306	12.482
Test	**Random Forest**	**7.452**	**10.558**

On the 2024 test period, Random Forest reduced MAE by **19.9%** and RMSE by **15.4%** relative to the historical baseline; validation results were mixed.



Volatility Feature Importance

Feature-importance interpretation: Validation-set permutation importance identified **VIX** as the most prominent model input. This indicates predictive dependence, not an independent causal effect; correlated features complicate attribution.



2024 Out-of-Sample Forecast Comparison

Forecast-chart interpretation: - **Stable periods:** Random Forest forecasts were comparatively smooth, but not consistently responsive to sudden volatility changes. - **August overprediction:** On **August 5, 2024**, the model forecast **61.66%** versus **14.60%** future realized volatility. With other inputs fixed, a local VIX scenario of **20 → 38.57** moved its prediction from **21.27% → 61.66%**. This is a model sensitivity test, **not causal**

identification. - **October–November spike:** JPM daily returns of **+11.54% on November 6** and **–4.32% on November 7** contributed to elevated subsequent 20-day realized-volatility windows. Overlapping targets can make a few extreme returns appear as an extended high-volatility region.

Forecast limitations: Only one test year was evaluated; adjacent 20-day target windows overlap. Extreme moves and data sources should be independently verified before making event-level claims.

5. Direct Pricing ML — Design, Not Yet Trained

Track	Prediction target	Current status
A — Volatility forecasting	Future 20D realized volatility	Preliminary Random Forest and importance analysis completed
B — Direct Chooser pricing	**Synthetic, frictionless Heston theoretical value**	Inputs/target defined; large-scale label generation and ML training pending

Direct-pricing inputs will include spot, strike, choice/expiry times, rates, dividend yield, and Heston parameters. Execution-adjusted prices can be studied as a *separate* target after agreeing on the market-friction scope. Synthetic labels assess how well ML approximates the assumed Heston model; they do **not** establish superiority against real-world option prices.

6. Mentor Feedback & Future Work

Tyler confirmed that static execution-cost adjustments are a valid component of market frictions. The current bid–ask spread, slippage, and transaction-fee analysis is therefore retained as the W04 extension.

Future extensions:

- **Dynamic Delta Hedging:** Evaluate cumulative rebalancing costs and hedging P&L under Heston-simulated price paths, with an explicit delta estimation method and hedging rule.
- **Heston Market Calibration:** Estimate stochastic-volatility parameters using observed option-market data rather than illustrative assumptions.

Immediate priority — W06–W07:

Develop a machine-learning surrogate for the **frictionless Heston theoretical Chooser price**. Generate synthetic pricing labels across varying contract and Heston parameters, then evaluate pricing errors, computational efficiency, and feature importance.

Execution-cost-adjusted pricing and dynamic hedging will remain separate extensions rather than initial ML targets.